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CTISum: A new benchmark dataset for Cyber Threat Intelligence summarization[☆]Wei Peng^{a,1}, Junmei Ding^{b,1}, Wei Wang^a, Lei Cui^a, Wei Cai^a, Zhiyu Hao^{a,*}, Xiaochun Yun^{a,*}^a Zhongguancun Laboratory, Beijing, China^b Beijing University of Posts and Telecommunications, Beijing, China

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ABSTRACT

Cyber Threat Intelligence (CTI) summarization involves generating concise and accurate highlights from web intelligence data with domain knowledge, which is critical to automatically summarize the knowledge and conclusion contained in CTI reports. Despite that, the development of efficient techniques for summarizing CTI reports, comprising facts, analytical insights, attack processes, and more, has been hindered by the lack of suitable datasets. To address this gap, we introduce CTISum, a new benchmark dataset designed for the CTI summarization task. Recognizing the significance of understanding attack processes, we also propose a novel fine-grained subtask: attack process summarization, which aims to help defenders assess risks, identify security gaps, and uncover vulnerabilities. Specifically, a multi-stage annotation pipeline is designed to collect and annotate CTI data from diverse web sources, alongside a comprehensive benchmarking of CTISum using both extractive, abstractive and LLMs-based summarization methods. Experimental results reveal that current state-of-the-art AI models (including GPT-4o) face significant challenges when applied to CTISum, highlighting that automatic summarization of CTI reports remains an open research problem. The code and example dataset can be made publicly available at <https://github.com/pengwei-ii/CTISum>.

1. Introduction

Cyber Threat Intelligence (CTI), also known as threat intelligence, is knowledge, skills and experience-based information concerning the occurrence and assessment of both cyber and physical threats as well as threat actors (Jo et al., 2022; of England, 2016). The CTI data often originates from diverse web sources, including forums, blogs, and open-source repositories, making it difficult for analysts to efficiently locate the most relevant and high-value intelligence from the web. Therefore, it becomes crucial to automatically summarize the knowledge contained in CTI reports, which could help analysts simply identify events, patterns, cyber attacks and conclusions. Furthermore, CTI summarization has broad applicability across domains like cybersecurity (Schatz et al., 2017), military intelligence (van Tilborg, 2005), technical alerts, threat modeling, and more, where complicated information needs to be distilled into concise insights.

With the ongoing advancement of summarization techniques, numerous datasets have emerged, such as CNN/DailyMail (Nallapati et al., 2016) and XSum (Narayan et al., 2018), among others. In addition,

the introduction of these datasets further drives the development of associated technologies like PGNet (See et al., 2017), BART (Lewis et al., 2020), BRIO (Liu et al., 2022), etc. Furthermore, domain-specific summarization researches have gradually arisen. Examples include biomedicine (Zhu et al., 2023), finance (Mukherjee et al., 2022), law (Shukla et al., 2022), etc. This further facilitates various applications involving aiding medical decision-making, generating financial reports, and summarizing legal documents. The boom of domain-specific summarization promotes the understanding of domain knowledge, terminology and context, which is crucial for generating high-quality summaries for further research.

Despite remarkable progress made in automatic text summarization, effective methods for summarizing CTI reports in the cybersecurity domain remain largely unexplored. This is primarily due to the lack of available dataset. The unique characteristics of CTI reports, such as technical jargon, evolving threat landscapes, longer reports and fragmented data information, make it challenging to create an annotation, which in turn hinders the development of robust models and evaluation benchmarks. However, recent advances in Large Language Models

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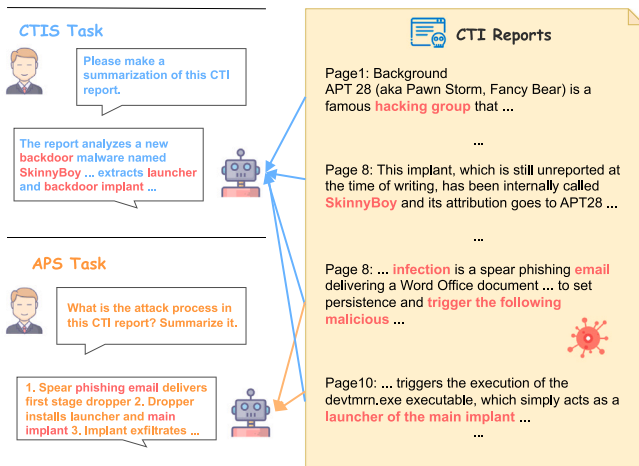


Fig. 1. An example in the CTISum. Blue arrow and orange arrow indicate the CTIS and APS tasks, respectively. Red font represents the key words. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(LLMs) (Hsiao and Collins, 2023; Cui et al., 2024; OpenAI, 2023; Touvron et al., 2023; A.I. Claude, 2023; A. Claude, 2023) have shown promising text understanding and generation capabilities. Therefore, this raises the question: *How might we construct a high-quality CTI summarization dataset with the assistance of LLMs?*

In this paper, we construct a new benchmark CTISum based on threat intelligence obtained from diverse web sources in the cybersecurity domain. In addition to the CTI Summarization (CTIS) task, considering the importance of attack process, we propose a novel fine-grained subtask, Attack Process Summarization (APS), to enable defenders to quickly understand reported attack behaviors, assess risks and identify security vulnerabilities. Specifically, a multi-stage annotation pipeline is designed to preprocess web CTI reports and obtain annotation data with the assistance of LLMs, which includes the data collection stage, parsing & cleaning stage, prompt schema stage and intelligence summarization stage. To further keep the high quality of the CTISum, we additionally employ expert reviews for double checking (details can be seen in Section 3.1). With proper data, automated systems can help synthesize lengthy CTI reports into concise and accurate summaries for different intelligence analysts. A sampled report and reference summary in CTISum are shown in Fig. 1. The system not only requires to make a general summary of the full document (task 1), but also needs the capability to focus on and generate the attack process (task 2). After cleaning and human checking, CTISum obtains 1371 documents and corresponding summaries. What makes CTISum a challenging dataset can be presented as follows. First, the average document length is about 2865 words. Representative prior approaches are incapable of processing documents surpassing 512/1024 tokens in length (after tokenization), such as GPT1(Radford et al., 2018), T5(Raffel et al., 2020), BART (Lewis et al., 2020) and so on. Although LLMs can deal with longer input, however, fine-tuning LLMs is resource consuming. Moreover, LLMs are generally pre-trained on generic datasets, so they often fall short when performing zero-shot tasks in specific domains, such as cybersecurity (it could be improved with domain knowledge). Second, the document-to-summary compression ratio on the two tasks are 14.32 (2865.60/200.04) and 22.23 (2865.60/118.27), respectively (details in Table 1). The high compression ratio shows competitiveness with the current long document summarization datasets, requiring systems to be extremely precise in capturing only the most relevant facts from lengthy documents in a minimal number of words. Finally, CTISum has another subtask, APS, which means the system should be capable of the ability to capture the fine-grained attack process described in CTI reports.

To demonstrate the challenges posed by the proposed CTISum dataset, we conduct comprehensive experiments comparing a range of extractive and abstractive summarization methods. The automatic and human evaluation indicate that existing approaches still have considerable limitations on the CTISum. The extractive methods struggle to identify salient information from lengthy and complex documents. They tend to produce incomplete summaries. Abstractive techniques face challenges in generating coherent and non-redundant summaries, as well as avoiding hallucinations. Both extractive and abstractive methods achieve low scores on automatic metrics like ROUGE. Our findings highlight the need for continued research to develop summarization techniques that can distill critical information in CTISum. The dataset and baselines will be released for further research.

The contributions can be summarized as follows:

- To the best of our knowledge, we make the first attempt to build a new benchmark CTISum with the CTIS task and a novel APS subtask in the cybersecurity domain, which focuses on summarizing the key facts or attack process from CTI reports.
- A multi-stage annotation pipeline is designed to obtain the high-quality dataset with the assistance of LLMs while manually controlling the quality, which consists of the data collection stage, parsing & cleaning stage, prompt schema stage and intelligence summarization stage.
- Comprehensive experiments on CTISum demonstrate the challenge of the proposed two tasks, meanwhile indicating a large space for future research.

2. Related work

2.1. Text summarization

A considerable amount of existing researches in automatic text summarization (Rush et al., 2015; Nallapati et al., 2016, 2017; Celikyilmaz et al., 2018; Lebanoff et al., 2019; Zhong et al., 2020) have explored to improve summary generation for news articles, using popular datasets such as CNN/DailyMail (Nallapati et al., 2016), XSum (Narayan et al., 2018), etc. In addition, scientific document summarization has emerged as another critical area, with datasets based on ArXiv/PubMed (Cohan et al., 2018), and other academic sources. The main approaches cover two aspects. First, extractive methods (Erkan and Radev, 2004; Cheng and Lapata, 2016; Zhou et al., 2018; Liu, 2019; Liu and Lapata, 2019; Zhong et al., 2020) involve identifying and extracting key sentences or passages from the original text to form the summary. For instance, Liu and Lapata (2019) introduce a document-level encoder based on BERT and propose a general framework for summarization. Second, abstractive techniques focus on generating new sentences to capture the meaning of the source text. For example, sequence-to-sequence models with attention (Luong et al., 2015; Nallapati et al., 2017; Peng et al., 2021) are usually utilized to generate the probability distributions on the vocabulary. In addition, See et al. (2017) present a hybrid pointer generator architecture for abstractive summarization. And more recently, transformer-based models like T5 (Raffel et al., 2020), BART (Lewis et al., 2020) and Longformer (Beltagy et al., 2020) have shown advanced performance on the summarization task.

Although current general summarization techniques have made some progress, their performance significantly decrease when transferred to CTI summarization tasks due to the lack of domain-specific fine-tuning data. The creation of the CTISum could provide an essential resource for domain adaptation, as well as developing and evaluating domain-specific summarization approaches.

2.2. Summarization methods

From the perspective of summarization generation strategy, text summarization models can be categorized into extraction-based and abstraction-based methods. Specifically, the two main paradigms are detailed as follows. Extractive summarization (Liu and Lapata, 2019; Zhong et al., 2020) focuses on selecting a subset of salient sentences or phrases from the original document to construct a concise summary. The core idea is to identify and extract the most informative content. For example, Liu and Lapata (2019) propose BERTSumExt, a neural extractive summarization model based on BERT. A classifier is then used for sentence selection to generate the final summary. Zhong et al. (2020) formulate extractive summarization as a semantic text matching task and develop MatchSum, a summary-level framework that generates multiple candidate summaries from the output of BERTSumExt and selects the one best matching the document as the final result. Abstractive summarization (Vaswani et al., 2017; Raffel et al., 2020; Lewis et al., 2020), in contrast, aims to generate novel sentences that semantically represent the key information. It leverages deep semantic understanding and natural language generation capabilities to produce coherent summaries. Vaswani et al. (2017) propose the Transformer architecture, a key milestone enabling abstractive summarization. Lewis et al. (2020) propose BART, an encoder-decoder language model based on a sequence-to-sequence Transformer architecture, achieving strong performance on various NLP tasks such as summarization and text generation after fine-tuning. Raffel et al. (2020) introduce T5, a unified text-to-text Transformer framework that reformulates all NLP tasks into text-to-text problems, where both input and output are natural language sequences. In addition, recent studies have focused on long document summarization, aiming to effectively capture and condense information from extended contextual inputs. Long document summarization (Beltagy et al., 2020; Touvron et al., 2023; OpenAI, 2023) aims to generate concise and coherent summaries from documents containing thousands of tokens. Representative approaches include: Beltagy et al. (2020) propose the Longformer, which adopts a sparse attention mechanism combining local sliding-window attention with global tokens to reduce the quadratic complexity of self-attention. This design enables efficient processing of sequences up to 16 K tokens. Touvron et al. (2023) introduce LLaMA2, a family of large language models trained on extended-context corpora with improved tokenizer efficiency and an optimized transformer depth-to-width ratio, which can extend up to 4096 tokens. According to OpenAI (OpenAI, 2023), GPT-4 supports a maximum context length of approximately 32,768 tokens, which makes it particularly effective for document-level summarization and multi-section content condensation tasks.

2.3. Domain-specific summarization

For the domain-specific summarization, there are many potential applications across different domains like government (Huang et al., 2021), finance (Mukherjee et al., 2022), court rulings (Shukla et al., 2022) and biomedicine (Zhu et al., 2023). For example, in the government domain, summarization systems can help process large volumes of regulatory text and legal documents. Huang et al. (2021) propose a novel efficient encoder-decoder attention with head-wise positional strides to effectively double the processed input sequence size. For the finance industry (Mukherjee et al., 2022), summarization of earnings reports, financial statements, and news can assist investors and analysts in making decisions. In addition, summarizing court rulings, case law, and litigation documents (Shukla et al., 2022) helps legal professionals in their work. Within biomedicine (Zhu et al., 2023), healthcare professionals can make informed decisions by using summaries of clinical trial findings, medical literature, and patient records. Different from the above domains, we make the first attempt to build the CTI summarization dataset in the cybersecurity domain, which helps fill the gap and enables new research into CTI summarization techniques.

2.4. Relation with CTI operational integration

Beyond summarization generation research, our CTISum benchmark has strong potential for integration into operational Cyber Threat Intelligence (CTI) pipelines, supporting automation and decision-making across multiple stages of the CTI lifecycle. (1) Integration with Threat Intelligence Platforms (TIPs). Threat Intelligence Platforms such as Malware Information Sharing Platform (MISP) and OpenCTI are central infrastructures for collecting, correlating, and sharing indicators of compromise (IOCs). Some studies (Wagner et al., 2016; Balasubramanian et al., 2025) emphasize the need for automated normalization and summarization of unstructured threat reports before ingestion into TIPs. The CTISum dataset directly supports this requirement: its CTIS task can automatically extract concise summaries of incidents and entities (threat groups, TTPs, IOCs) that can be fed into systems like MISP or OpenCTI for enrichment and correlation. (2) Support for threat intelligence automation and fusion. Operational CTI automation frameworks—such as TheHive-Cortex, MISP connectors, and TIP orchestration tools—increasingly rely on machine-readable intelligence. CTISum's Attack Process Summarization (APS) task provides a fine-grained representation of adversary behaviors and sequences of TTPs, which can be aligned with ATT&CK techniques. This mapping enables automated workflows where APS outputs can populate ATT&CK technique fields (e.g., T1059 – Command and Scripting Interpreter), allowing TIPs or SOAR systems to automatically trigger playbooks for detection rule generation or alert triage. (3) Connection to Adversary Emulation Frameworks. Frameworks such as CALDERA, Atomic Red Team, and ATT&CK Navigator are used for adversary emulation and red-team planning. Recent work (Büchel et al., 2025; Rani et al., 2024) explored converting textual CTI into machine-actionable TTP sequences for automated adversary simulation. CTISum's APS summaries naturally fit into this process, as they provide structured, succinct descriptions of attack process that can be programmatically mapped to ATT&CK techniques and imported into CALDERA agents for emulation. Thus, CTISum offers a bridge between unstructured CTI narratives and automated adversary modeling. Future work will explore fine-tuning LLMs on CTISum to generate structured, STIX-compatible summaries and to design plug-ins for MISP or CALDERA that demonstrate this integration in practice. In a typical deployment scenario, CTISum can be positioned downstream of raw threat intelligence ingestion pipelines, where CTI reports are automatically summarized into concise outputs. These summaries can further be transformed into structured representations compliant with widely adopted standards such as STIX (Structured Threat Information Expression). For example, entities extracted or verified during summarization — such as threat actors, indicators of compromise (IOCs), attack patterns, and vulnerabilities — can be mapped to corresponding STIX objects and relationships, enabling export to threat intelligence platforms or SIEM systems.

3. Data construction

As shown in Fig. 2, the proposed multi-stage annotation pipeline consists of four stages. The first stage is collecting data from diverse web sources, namely open source report (CyberMonitor et al., 2019), threat encyclopedia, and security company report to ensure the diversity and coverage. The second stage is parsing and cleaning which aims to parse original PDF reports into readable text, then obtain high-quality input data by manually cleaning and filtering. Next follows the prompt schema stage, where domain experts design various prompts based on different tasks, to fully utilize the powerful text generation capabilities of LLMs and prepare for the annotation. Finally, the intelligence summarization stage. To combine system productivity and expert experience efficiently, we transform the manual annotation process into ranking and classifying, having experts review and rank the outputs of LLMs to obtain the final summary. Next, we first introduce the multi-stage annotation pipeline, and then make a statistics and analysis of the CTISum.

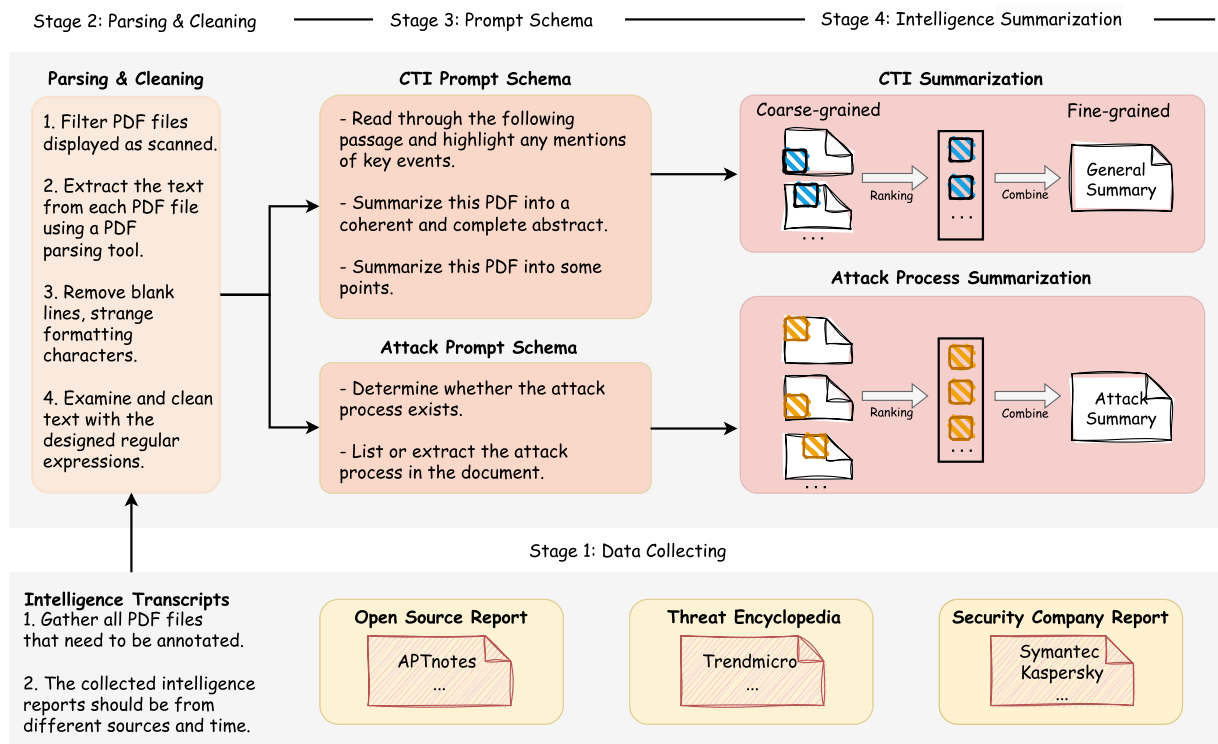


Fig. 2. The overview of the proposed multi-stage annotation pipeline for two tasks.

3.1. The multi-stage annotation pipeline

The multi-stage annotation pipeline focuses on obtaining the high-quality CTISum dataset, where we attempt to combine the strengths of both expert experience and LLMs (like GPT-4 (OpenAI, 2023) or Claude2 or 3.5 (A. Claude, 2023)) to achieve semi-automatic summarization annotation for efficiency in the cybersecurity domain.

Stage 1: Data collecting. To gather relevant CTI reports and build CTISum, we use the reports scraped from diverse web sources, including web open source report like APTnotes,² threat encyclopedia like Trendmicro,³ security company report like Symantec⁴ and various cyber threat groups like Carbanak, OilRig, Sandworm⁵ to ensure the diversity and coverage of the CTISum. The range of collected data is covered from 2016 to 2025. In addition, CTISum also provides extra new and up-to-date CTI reports (about 120), including dark web reports, industry reports, country-specific threat landscape reports and timely analyses of emerging cybersecurity trends reports. These data sources can serve as additional pretraining corpora for future research, enabling models to acquire richer domain knowledge about APT and real-world attack scenarios. The criteria is that these documents are continuously maintained by the community and they are widely utilized in the CTI field. Moreover, we select these CTI documents whose pages are only from reputable blogs, vendor sites, famous github community or government CSIRT feeds. And they should contain the following rules: (i) English language, (ii) technical detail on malware/TTPs, (iii) analytical conclusions, and (iv) non-redundancy after deduplication. For the summarization tasks, a balanced dataset of 1371 documents is sampled and annotated with two categories: CTIS and APS.

Stage 2: Parsing and cleaning. In order to obtain high-quality input data, the second stage focuses more on parsing and further cleaning the parsed text.

Given that CTI reports are currently formatted as PDF files, which have complex structure and cannot be directly used as the input to the system, one necessary step is to parse the PDFs and extract readable text. Specifically, the gathered PDFs are firstly filtered to only include those containing actual text content, rather than scanned documents which cannot be parsed by certain tools like PDFMiner. Then, an off-the-shelf PDF parsing library⁶ is leveraged to programmatically extract the readable text from each document, while figures are discarded. Finally, the output text is saved in TXT format, allowing transferability to downstream systems.

After parsing, it is indispensable to clean and filter the extracted text for high-quality input data to the LLMs. This post-processing helps improve the quality and consistency of the textual data. One methodology is to manually review samples of the extracted text to identify issues that need to be cleaned. Some rules are:

- Correcting encoding issues when parsing error.
- Removing blank lines and odd characters that get extracted but provide no value.
- Cleaning up irregular text like ellipses and page numbers that are extracted from tables of contents, lists, etc.
- Removing non-English and useless characters.
- Deduplicating and removing strings like long sequences of consecutive IP addresses, hashes, etc.
- Filtering with regular expressions. Details in Appendix A.2.

The parsing and cleaning stage ensures higher quality input text for the next stage of analysis and annotation.

² <https://github.com/kbandla/APTnotes>.

³ <https://www.trendmicro.com>.

⁴ <https://www.broadcom.com/support/security-center>.

⁵ <https://github.com/dessertlab/CTI-HAL>.

⁶ <https://github.com/euske/pdfminer/tree/master>.

Stage 3: Prompt schema. The purpose of this stage is to develop a detailed prompt schema to guide the LLMs (Claude2 or 3.5, ChatGLM3, etc.) in producing candidate summaries, such as highlighting key events and summarizing reports, attack processes, some useful annotations, etc. To generate accurate and consistent summaries for the CTIS task and the APS task, CTI experts analyze the parsed and cleaned text and then design the CTI prompt schema and attack process prompt schema, respectively. As shown in Fig. 2, several key annotation prompts are defined as follows:

- **Key Events:** The instruction, **read through the following passage and highlight any mentions of key events**, will focus on important cyber threat events like new malware campaigns, critical vulnerabilities, or notable hacks.
- **Coherent Summary:** To obtain the complete and coherent abstract, the instruction, **summarize this PDF into a coherent and complete abstract**, is leveraged.
- **Additional Points:** The purpose of the instruction **summarize this PDF into some points** is to generate some additional aspects for supplementary.
- **Determination:** Considering some reports do not contain the attack process, the instruction **determine whether the attack process exists** is designed.
- **Attack Process:** **List or extract the attack process in the document** is utilized for the APS task.

The prompt schema outlines each annotation prompt in detail, which is a critical component that will guide the LLMs to produce useful, accurate, and consistent summaries.

Stage 4: Intelligence summarization. To combine system productivity and expert experience more efficiently, we transform the manual annotation process into ranking and classifying model-generated results, having three experts review and rank the outputs into overall summaries. Specifically, LLMs like GPT-4o (OpenAI, 2023), Claude2 or 3.5 (A. Claude, 2023), ChatGLM3 (Zeng et al., 2022) are leveraged to generate multiple coarse-grained summaries for the proposed two tasks. First, we employ proprietary LLMs such as GPT-4o and Claude 3.5 due to their exceptional superior factuality and narrative coherence when processing CTI scenarios, which often involve intricate, high-stakes details requiring reliable output. For the diverse of annotation phase, open-source models like ChatGLM3 are also utilized, however, these models exhibit significantly higher hallucination rates — generating inaccurate or fabricated information — and demonstrate poorer comprehension of CTI content, leading to summaries that are frequently less fluent and logically inconsistent. We randomly select 100 CTI reports and compare candidate summaries generated by proprietary models (GPT-4o) and open-source models (ChatGLM3). The results indicate that ChatGLM3 exhibited hallucinations at a rate 42% higher than GPT-4o. Hallucination discussion can be seen in Appendix A.3. To avoid overgeneralization, we explicitly clarify that this observation is empirical and limited to our experimental conditions, prompts, and dataset, rather than a universal claim about all open-source or proprietary models. In fact, domain experts usually leverage the candidate summaries from proprietary LLMs.

Then, three domain experts are involved in reviewing and ranking the candidate summaries, discussing and selecting the better content to combine into a final gold summary. If the generated summaries turn out to be inadequate, the example would be discarded. Criteria for judging inadequate summary can be seen in the following, a candidate is removed if it violated any of the following:

- **Factual hallucination:** Entity correctness (e.g., threat group, malware family, CVE ID). Verification of TTP descriptions against the source document.
- **Completeness:** Discard for fabricated IOCs or invented attack steps. Miss larger than 2 consecutive attack steps that appear in the ground-truth timeline.

Table 1

Statistics of CTISum dataset and existing different domain summarization datasets. Doc.: document, Avg.: average, Len.: length.

Dataset	# Doc.	Avg. Doc. Len.	Avg. Sum. Len.	Domain
General domain				
ARXIV/PUBMED (Cohan et al., 2018) ^a	346,187	5179.22	257.44	Academic
CNN (Nallapati et al., 2016) ^b	92,579	760.50	45.70	News
DAILYMAIL (Nallapati et al., 2016) ^b	219,506	653.33	54.65	News
XSUM (Narayan et al., 2018) ^b	226,711	431.07	23.26	News
BOOKSUM chapters (Mukherjee et al., 2022) ^a	12,630	5101.88	505.42	Books
Specific domain				
GOVREPORT (Huang et al., 2021)	19,466	9409.40	553.40	Government
DISCHARGE (Zhu et al., 2023) ^b	50,000	2162.29	28.84	Biomedicine
ECHO (Zhu et al., 2023)	162,000	315.30	49.99	Biomedicine
ECTSUM (Mukherjee et al., 2022)	2425	2916.44	49.23	Finance
IN-EXT (Shukla et al., 2022)	50	5389	1670	Court rulings
UK-ABS (Shukla et al., 2022)	793	14,296	1573	Court rulings
CTISUM	1371	2865.60	200.04	Cybersecurity

^a The numbers for the datasets are copied from the paper (Mukherjee et al., 2022).

^b The ones are copied from the work (Narayan et al., 2018).

- **Redundancy:** Any 2-sentence overlap (larger than 10-gram) with each other or with the source.
- **Security-tone violation:** Use speculative language (may, possibly) for events that the source states as fact.

After LLMs generation, 92 of the remaining 1463 documents (6.28%) are dropped because no candidate summary meets the adequacy criteria described above. Consequently, the public release contains 1371 documents. The domain experts play a key role in evaluating and improving the model-generated summaries to produce a high-quality benchmark. Their human judgment and domain expertise complement the capabilities of LLMs. This collaborative human-AI approach allows the creation of abstractive summaries that are coherent, relevant, and accurate.

3.2. Annotation criterion

To ensure consistency across annotators, we implement a multi-round calibration and validation protocol prior to large-scale annotation.

(1) **Calibration phase.** Before annotation began, all participating annotators discuss joint calibration session. The session uses 30 sample CTI reports representing diverse threat types (APT campaigns, ransomware incidents, vulnerability analyses). Annotators first generate independent summaries under both CTIS and APS schemas, then participate in structured group discussions to discuss differences and clarify task-specific interpretation rules (e.g., defining what constitutes a “complete attack step” or “analytic conclusion”). It guarantees the fact for judging factual completeness, technical relevance, and writing style consistency.

(2) **Iterative agreement monitoring and criterion rules.** During the initial annotation phase, we continuously measured inter-annotator agreement on 30 selected samples about three times. Disagreements are analyzed daily to identify systematic biases, after which updated

examples are added to the annotation pool. Annotators evaluate each candidate based on the predefined criteria.

(3) Dual validation stage. For the final release, two CTI experts independently re-examine a 15% random subset of summaries to validate factual correctness, attack-chain completeness, and adherence to the prompt schema. Discrepancies are adjudicated through consensus meetings, ensuring high factual reliability in the published dataset. This process is repeated (about two rounds) until the sampled final summaries reached a stability point, where over 99% of them satisfied all four criteria simultaneously. From this validation stage, we draw two main conclusions: first, the majority of retained summaries are judged to be factually correct under our strict criteria; second, the remaining errors are primarily minor omissions rather than critical factual mistakes.

3.3. Data statistics

Table 1 describes the statistics of the CTISum dataset and existing summarization datasets from different domains. The datasets are divided into general and specific domain categories, with the latter attracting substantial interest in recent years from 2021 to 2024. And we make the first attempt to build the CTI summarization dataset in the cybersecurity domain. Specifically, CTISum has 1371 documents from diverse web sources, with an average document length of 2865 words, with an average CTI summary length of 200 words, and with a high compression ratio of 14.32. Notably, CTISum is the only dataset with the subtask, while other datasets simply make a general summary of the document. The average length of the attack process summaries is about 118 words, which is shorter than the general summaries. This highlights the challenging nature of generating focused summaries that capturing the attack process from long documents. In addition, the Fleiss Kappa evaluation of the dataset can be seen in Section 4.6.

In summary, the table highlights that CTISum is a challenging dataset for summarization in the cybersecurity domain, with a novel subtask (attack process summarization) and high abstraction requirements. The CTISum dataset helps fill the gap and enables new system design in future work.

4. Experiments

4.1. Task formulation

Existing approaches usually define the summarization task as a sequence-to-sequence task. Specifically, the problem formulation can be formulated as follows. Given a document, the model should generate the summary that is concise, coherent and consistent. For the proposed CTISum, there are two tasks, including CTIS task and APS task. For CTIS task, the object is to output a general summary of the CTI report. Similarly, the object of the APS task is to generate a summary of the attack process.

4.2. Evaluation metrics

The automatic and human A/B evaluation are considered to validate the performance of the current SOTA models.

Following papers (Hsu et al., 2022; Mukherjee et al., 2022), BERTScore (Zhang et al., 2020) and ROUGE- n (R- n) (Lin, 2004) are taken as evaluation metrics, which are widely used for evaluating the quality of summarization. BERTScore is a metric for evaluating text generation models, which measures the similarity between the generated text and reference text using contextual embeddings from pre-trained BERT. ROUGE- n refers to the overlap of n -grams between the generated and reference summaries, which is used as an auxiliary metric. ROUGE-1 evaluates unigram overlap, ROUGE-2 measures bi-gram overlap, and ROUGE- L calculates longest common subsequence overlap. These metrics compare matching units such as words and

phrases between the generated and reference summaries, with higher scores indicating better summarization quality. ROUGE- L generally corresponds best with human judgments of summary quality.

In previous studies, human evaluation is usually conducted by crowdsourcing workers who rate responses on a scale from 1 to 5 from the aspects of correctness, relevancy, etc. However, the criteria can vary widely between different individuals. Therefore, this study adopts the human A/B evaluation for a high inter-annotator agreement. Given the generated summaries of two models A and B, analysts are prompted to go through an entire CTI report and choose the better one for each of the 80 randomly sub-sampled test instances. To ensure objectivity and domain adequacy, we employ 4 annotators with complementary expertise:

- Three cybersecurity assistant researcher familiar with malware behavior and report analysis.
- One senior CTI analysts (over five years of experience in threat analysis, incident response, and MISP/ATT&CK workflows).

All annotators are briefed with a standardized evaluation protocol prior to annotation. They are provided with detailed definitions and examples for each evaluation dimension:

- Factual Correctness: whether statements in the summary are verifiable from the source CTI report and contain less hallucinated entities, CVEs, or TTPs.
- Relevance: whether the summary captures the key threat events, actor names, and attack vectors relevant to the report's focus.
- Coverage: whether the summary includes all critical elements of the described attack process, particularly for the APS task.
- Entity-level Correctness Metric (ECM): whether the named entities (e.g., threat actors, malware families, vulnerabilities, organizations, and attack techniques) contained in the generated summaries are factually correct and consistent with the source.

In cases of disagreement, a majority voting rule is applied; if all three disagreed, the senior analyst acted as a tiebreaker.

The final results are determined by majority voting. If the three annotators reach different conclusions, the fourth annotator will be brought in.

4.3. Experimental setting

The implementation of baselines is based on the HuggingFace framework. The AdamW optimizer (Loshchilov and Hutter, 2017) with $\beta_1 = 0.9$ and $\beta_2 = 0.99$ is used for training, with an initial learning rate of $3e-5$ and a linear warmup with 100 steps. The batch size is set to 16 for training, and we use a batch size of 1 and a maximum of 128 decoding steps during inference. Top- p sampling is set to 0.9, temperature $\tau = 0.7$. The epoch is set to 5. For preprocessing, we randomly split the dataset into train, validation and test set with a ratio of 8:1:1.

4.4. Baselines

Several SOTA approaches are illustrated for comparison. The models can be mainly divided into extraction-based model, abstraction-based and long-document-based methods.

Extractive model. Extractive summarization involves selecting a subset of salient sentences, phrases, or words from the original document to form the summary. The key idea is to identify and extract the most important content, and then arrange extracted content to flow logically. We simply introduce MatchSum as follows.

MatchSum (Zhong et al., 2020) formulates the extractive summarization task as a semantic text matching problem and develops a novel summary-level framework MatchSum, which generates a collection of possible candidate summaries from the output of BertSumEXT. The candidate that matches best with the document is selected as the final summarized version.

Table 2

Performance of automatic evaluation on CTIS and APS tasks. LLAMA2 is in the zero-shot setting because of the lack of resources. The best results are highlighted in **bold**.

Model	CTIS validation				CTIS test			
	BERTScore↑	R-1↑	R-2↑	R-L↑	BERTScore↑	R-1↑	R-2↑	R-L↑
Extractive								
MatchSum (Zhong et al., 2020)	83.91	39.31	13.39	20.60	83.75	33.76	9.34	18.91
Abstractive								
Transformer (Vaswani et al., 2017)	45.83	34.15	12.05	19.88	45.17	32.94	10.37	18.55
T5 (Raffel et al., 2020)	70.48	43.75	16.58	28.06	69.21	42.41	14.35	27.32
BART-base (Lewis et al., 2020)	70.36	44.45	16.26	27.24	69.24	42.48	14.30	26.23
BART-large (Lewis et al., 2020)	71.21	47.11	18.32	29.20	70.41	45.76	16.88	29.03
Long-document-based								
Longformer (Beltagy et al., 2020)	71.06	46.97	17.21	28.22	70.31	45.39	15.66	27.09
LLAMA2 (Touvron et al., 2023) (zero-shot)	33.87	21.35	5.93	20.51	32.15	20.89	5.66	19.22
GPT-4o (zero-shot)	65.82	40.18	13.22	24.33	64.31	38.70	12.71	22.98
Model	APS validation				APS test			
	BERTScore↑	R-1↑	R-2↑	R-L↑	BERTScore↑	R-1↑	R-2↑	R-L↑
Extractive								
MatchSum (Zhong et al., 2020)	83.93	33.86	9.86	19.17	83.75	33.73	9.35	18.88
Abstractive								
Transformer (Vaswani et al., 2017)	40.43	28.22	7.86	16.50	39.15	27.03	6.21	15.48
T5 (Raffel et al., 2020)	68.41	36.20	10.27	24.70	66.06	32.09	7.62	21.78
BART-base (Lewis et al., 2020)	70.47	40.47	12.33	25.28	70.70	39.42	11.13	24.45
BART-large (Lewis et al., 2020)	71.31	41.88	12.76	26.54	70.32	40.25	11.65	25.06
Long-document-based								
Longformer (Beltagy et al., 2020)	70.98	41.35	11.77	25.36	70.41	39.91	9.78	23.93
LLAMA2 (Touvron et al., 2023) (zero-shot)	21.06	15.68	2.23	13.14	20.38	14.66	2.02	12.28
GPT-4o (zero-shot)	48.77	32.06	9.11	18.69	45.39	30.55	8.61	17.75

Abstractive model. Abstractive summarization involves generating new phrases and sentences that convey the most important information from the source document, which uses semantic understanding and language generation technology to make a summarization. Hence, we fine-tune BART (Lewis et al., 2020) and T5 (Raffel et al., 2020) from the HuggingFace library.

BART (Lewis et al., 2020) is an encoder-decoder language model based on a sequence-to-sequence architecture, which achieves strong performance on a variety of NLP tasks like summarization, question answering and text generation after fine-tuning.

T5 (Raffel et al., 2020) uses a standard Transformer encoder-decoder, which converts all NLP tasks into a unified text-to-text format where the input and output are always text strings. T5 comes in several sizes including T5-Small, T5-Base, T5-Large, etc. In this paper, the setting is based on T5-Base.

Long-document-based model. Long-document-based summarization refers to generating summaries for documents that are significantly longer than typical texts, e.g. 1024 tokens, like Longformer (Beltagy et al., 2020), LLAMA2 (Touvron et al., 2023), etc.

Longformer (Beltagy et al., 2020) is an extension to the Transformer architecture, which leverages an attention mechanism whose computational complexity grows linearly as the length of the input sequence increases, making it easy to process thousands of tokens or more. LLAMA2 (Touvron et al., 2023) is a Transformer-based large language model, which is trained on massive text datasets to learn natural language patterns and generate human-like text. The context length of LLAMA2 can extend up to 4096 tokens, allowing it to understand and generate longer texts. In this paper, the 7 billion LLAMA2 is utilized to compare against fine-tuned models in zero-shot setting. GPT-4o (OpenAI, 2023) is a multimodal, multilingual generative LLM released by OpenAI. Its context length has reached 128K tokens, making GPT-4o perform better when processing long text or complex documents.

4.5. Main results

Automatic evaluation. We compare the performance of extractive models, abstractive models and long-document-based models on CTIS and

APS tasks. As depicted in Table 2, extractive models perform the almost worst on both tasks, demonstrating that intelligence summarization in cybersecurity involves more than just extracting sentences. Interestingly, they obtain the best BERTScore, one possible reason is that these two models are based on BERT, leading to high BERTScore. Compared with MatchSum, abstractive model like BART-large (about 374M) achieves better result, about 10.12% and 6.18% gain on ROUGE-L on two tasks, which shows that the abstractive models are more suitable for these two tasks. Then, long-document-based model like Longformer (102M) obtains similar results compared with BART-base (about 121M). Notably, LLAMA2-7B and GPT-4o (zero-shot) do not attain further improvement on all the evaluation metrics, leaving noticeable room for future work on domain LLMs fine-tuning and few-shot learning on the benchmark. Finally, we find that performance is worse on the APS task, likely because the attack process exhibits properties like variability, concealment and complexity, making APS more challenging than general summarization. Enhancing models with domain knowledge for APS task remains an important direction for future research.

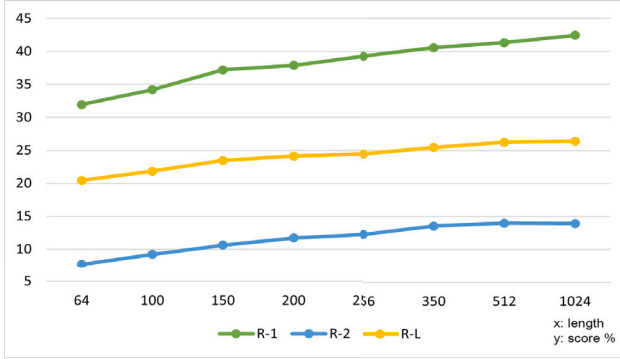
We explain that ROUGE can reward summaries that are lexically similar to the reference while still containing factual errors, such as incorrect threat actor names or misattributed vulnerabilities. This issue is particularly pronounced in CTI, where correctness of entities, temporal information, and attack sequences is more critical than surface-level wording similarity. We therefore position ROUGE as the coarse, auxiliary metric used mainly for comparability with prior summarization literature.

Human A/B evaluation. In addition to the automatic evaluation, human A/B evaluation (Peng et al., 2023) is conducted to validate the effectiveness of current SOTA models. In this section, the results of BART-large and Longformer are compared. We do not include extractive models in our comparison. This is because summaries generated by extractive models consist entirely of sentences directly copied from the source text. The results in Table 3 demonstrate the consistent conclusion with automatic evaluation. It can be seen that the summaries from BART-large are much more preferred than those of the baselines in terms of the three aspects. For example, compared with Longformer,

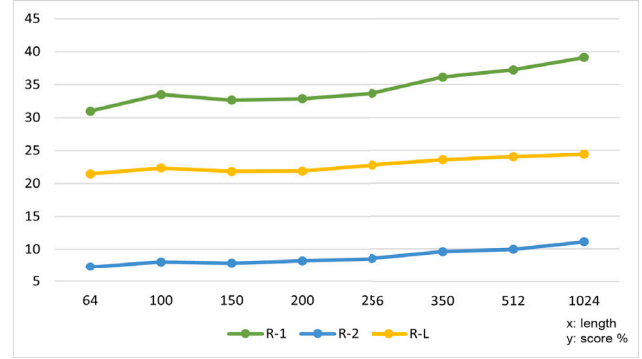
Table 3

Results for the human evaluation of model-generated summaries by three intelligent analysts.

Model	CTIS task				APS task			
	Correctness	Relevance	Coverage	ECM	Correctness	Relevance	Coverage	ECM
BART better	33.33	26.67	46.67	60.00	40.00	33.33	46.67	55.33
Longformer better	26.67	20.00	20.00	34.67	26.67	26.67	20.00	41.33
Both equally good	40.00	53.33	33.33	5.33	33.33	40.00	33.33	3.33



(a) CTIS task.



(b) APS task.

Fig. 3. Input length analysis of two tasks.**Table 4**

Fleiss Kappa evaluation of CTISum dataset.

Score	1	2	3	4	5	Fleiss Kappa
Result	0	9	58	129	54	0.37
Score	1 and 2		3	4 and 5		Fleiss Kappa
Result	9		58	183		0.61

BART is superior in terms of the coverage metric, indicating that the larger the number of parameters, the better the generated summaries. Besides, it is noted that BART does not significantly outperform Longformer in the relevance metric, and this is probably attributed to the powerful understanding ability of the PLMs, but BART still obtains decent improvements.

4.6. Fleiss Kappa analysis

To verify the consistency of the final summary generation quality, we select 5 experts to execute a consistency evaluation of 50 sampled summaries using Fleiss Kappa, where the scores are rated from 1 to 5, with 1 very negative, 2 negative, 3 neutral, 4 positive, and 5 very positive. As shown in Table 4, the final Fleiss Kappa score is 0.37, indicating relatively low agreement. This can be attributed to the fact that summary generation belongs to language generation task, where different experts tend to have subjective views on summary quality (e.g., examples that annotated as positive or very positive are inconsistent, but they are all good), which differs greatly from the objectivity of classification tasks. However, we also analyze the score distribution of all experts, among the total 250 expert ratings (50 documents $\times$ 5 raters = 250 scores), 241 received a score of 3 or higher (96.4%), demonstrating that the quality of the annotated dataset is controllable and accurate. Moreover, we have conducted another round of Fleiss Kappa analysis on the previous annotation scores from multiple raters. Specifically, we combine score 1 and 2 into one bin, keep score 3 as a separate bin, and combine score 4 and 5 into another bin. This result in a Fleiss Kappa score of 0.613184, which indicates the consistency evaluation.

4.7. Input length analysis

In this section, the input length analysis is performed to study the model's sensitivity to input length. We evaluate the impact of different input lengths on the BART-base on CTI and APS tasks, which can accommodate computational constraints. The goal using BART-base is to analyze scaling trends rather than absolute performance. We expect these trends to largely hold for BART-large as well, given that BART-large shares the same architecture but with increased capacity. However, the magnitude of improvements may differ on other PLMs or LLMs. Specifically, we divide the different length to get better insight into the current models. As shown in Fig. 3, where the x-axis represents the input length and the y-axis denotes the metric value. The conclusions can be drawn: (1) As the input length increases, the model performance improves steadily, highlighting the model's ability to process and summarize longer sequences. (2) As the length of BART reaches 1024 tokens, performance on our tasks continues to improve. To further investigate the impact of longer length, we increase the size of Longformer and find that lengthening to 3500 tokens (75th percentile of length distribution) results in optimal performance, which suggests the advantage of long-distance modeling. (3) The model achieves poorer performance on the APS task compared to the CTIS task, indicating it struggles more with summarizing technical attack details than threat intelligence reports. In summary, the experiments provide insight into current models' ability to capture long-distance dependencies in documents and summarize essential information as the input size varies. Further work could focus on improving the model's ability of length to boost performance on lengthy, technical summarization tasks.

4.8. Few-shot analysis

Analyzing the impact of data size can help to understand the relationship between data size and summarizing performance. To demonstrate such a relationship, in this section, few-shot analysis experiments on BART-base are conducted to study model's sensitivity in different training data sizes. We experiment with 10%, 25%, 40%, 55%, 70%, and 100% of the total training data to analyze the impact on CTIS task and APS task. The experimental results are shown in Fig. 4 with

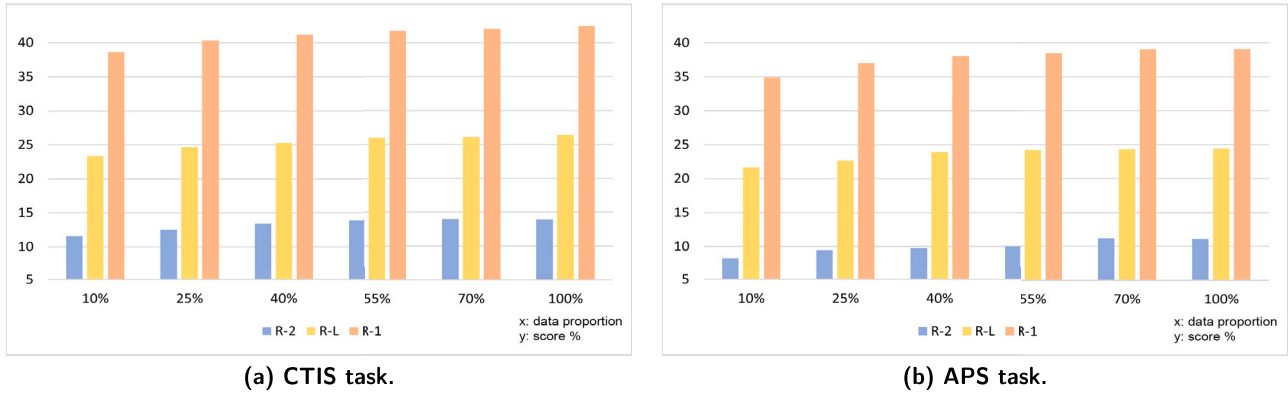


Fig. 4. Few shot experiments of the proposed two tasks.

Table 5

Case analysis on CTIS and APS tasks. **Orange words** mean the hallucination generated by the baseline. **Blue words** present the correct result. **Red words** indicate some of the key information that needs to be extracted.

Task	CTIS task
Document	ShadowGate returns to worldwide operations With evolved Greenflash Sundown ... After almost two years of sporadic restricted activity, the ShadowGate campaign has started delivering cryptocurrency miners with a newlyupgraded version of the Greenflash Sundown exploit kit ... ShadowGate was active since 2015 but restricted operations after a takedown in 2016 ... In April 2018 , ... visitors were redirected to the Greenflash Sundown exploit kit and infected with a Monero cryptocurrency miner .
BART	The PDF describes the activities of a threat actor group called ShadowGate ... The campaign has been actively evolving its exploit kit since 2016 ... ShadowGate started distributing cryptocurrency miners using a new version of Green Sundown , likely to avoid ... The Monero and Zcash mining were used ...
Ground truth	The report analyzes the resurgence of a cybercrime campaign called ShadowGate, which is using an upgraded version of its custom exploit kit, Greenflash Sundown, to infect victims worldwide with cryptocurrency mining malware. ShadowGate was active since 2015 but restricted operations after a takedown in 2016 . It recently resumed attacks ... Monero cryptocurrency miners after infection ...
Task	APS task
Document	Staring at a major financial resource crunch, MSME's are worst affected due to the ongoing COVID-19 pandemic. We observed one similar wave on MSME in late April 2020 — it was a phishing campaign luring victims with COVID-19 themed maldocs ... Victims received an email with attached zip “face mask order.zip” which contained the aforementioned maldoc ... The attack begins after opening “face mask order.doc” . This RTF is weaponized with exploit which triggers CVE-2017-11882 vulnerability to execute arbitrary code ... injected Agent Tesla payload in RegAsm.exe ...
BART	1. The phishing email contains a malicious RTF file titled “face mask order.zip” 2. The RTF contains an exploit that triggers CVE-2017-11882 to execute arbitrary code. 3. The exploit extracts the OLE objects from RTF files. 4. The OLE object contains a VBScript script that executes an Equation Editor payload. 5. The Equated Editor payload connects to a command and control server to receive commands. 6. Command server sends commands to the C2 server to execute additional payloads. 7. Additional payloads are delivered to the infected ...
Ground truth	1. Gorgon APT sends phishing emails with COVID-19 themed Word documents to targets in the MSME sector. 2. The documents contain an exploit for CVE-2017-11882 to execute malicious code. 3. The code drops a visual basic script named ServerCrypted.vbs . 4. The VBScript executes a PowerShell command to download additional payloads. 5. The first payload is an injector DLL. 6. The second payload is the Agent Tesla Remote Access Trojan. 7. The injector DLL loads itself into memory using PowerShell. 8. Agent Tesla is injected into the RegAsm.exe process. 9. Agent Tesla ...

several conclusions: (1) Performance of BART-base improves as the training data size increases, indicating that model quality benefits from more abundant training data. Furthermore, the result in the figure is still growing, which suggests further improvements with larger training data size. (2) Performance on the APS task is worse than the CTIS task, especially in few-shot setting (e.g., 10%, 25%). Therefore, how to improve the performance of summarization models in few-shot setting for cybersecurity domain will be an important future research track. And data augmentation with domain knowledge is also another area worth researching.

4.9. Case analysis

Qualitatively, we observe interesting examples on the BART-base from CTIS and APS tasks in Table 5. In case one on CTIS task, the model

generates hallucination about time **since 2016**, leading to incorrect time information in the summary. This phenomenon likely occurs because if the source document contains multiple time details, the model may confuse about which time points are being referenced. Therefore, addressing this challenge will be critical for future improvements. In case two on APS task, while generated attack process summary can include some factual information **CVE-2017-11882**, many factual inconsistencies errors exist, such as **OLE** and **Equation Editor**. This stems from attack behaviors that exhibit properties of complexity, diversity, and contextual dependence, which makes the APS task difficult. Moreover, by closely examining the examples in Table 5, several concrete issues become evident in both CTIS and APS tasks. The CTIS case shows multiple forms of factual inconsistency and hallucination. (1) Temporal error: The source document states the campaign has been active since

2015, whereas the BART-generated summary incorrectly reports since 2016, representing a clear temporal hallucination. (2) Entity error: The original report identifies Monero cryptocurrency miner, but BART replaces it with Zcash mining, a fabricated entity inconsistent with the source. This is a clear case of entity hallucination, likely caused by model overgeneralization from frequent cryptocurrency-malware associations seen in training. These CTIS-specific patterns support our measured finding that CTIS suffers higher hallucination rates with the current models, especially fabricated CVEs, malware families, and threat actors. This originates primarily from the combination of (1) similar surface-level wording across CTI reports, which encourages models overgeneralization, and (2) the high compression ratio, which pressures the model to invent summary.

As for the APS task errors, the APS example exhibits a different pattern. (1) Missing procedural steps: The ground truth contains nine steps, but BART's output includes only seven, omitting some steps. (2) Loss of specificity: Technical operations such as "deploying encoded PowerShell loader" are frequently replaced with vague statements like "the attacker executed commands", representing cross-section omission due to long-document fragmentation. These observations align with our broader statistics: APS suffers significantly higher omission rates, due to the intrinsic multi-step procedural structure. Unlike CTIS, APS requires maintaining sequential logic and fine-grained behavioral transitions, which seq-to-seq models struggle to preserve without explicit structural supervision.

The different error tendencies of CTIS and APS suggest that future research may require task-specific mitigation strategies. For CTIS task, hallucination suppression, entity grounding, temporal consistency checks are important research direction. For APS task, structured extraction, chain-of-thought reasoning for step sequences, and explicit TTP ordering mechanisms, too. Overall, these cases highlight important limitations, but also provide opportunities to advance CTIS and APS tasks through innovations targeting core summarize-ability challenges in the cybersecurity domain.

5. Conclusion

With a lack of publicly available intelligence data, in this paper, we make the first attempt to propose CTISum benchmark which provides a valuable resource to spur innovation in this critical but unobserved cybersecurity domain. The creation of this intelligence-focused summarization benchmark represents an important step toward developing AI systems that can effectively process and synthesize CTI reports. To combine system productivity and expert experience more efficiently, a multi-stage annotation pipeline is designed for obtaining the high-quality dataset. Our experiments reveal that while current models can produce decent summaries, there is significant room for improvement in capturing key details accurately. The technical cybersecurity terminology and complex contextual concepts present in the reports pose great challenges for generating intelligence summaries. In the future work, alleviating domain-tuned hallucinations is especially critical for CTIS, as it restricts the invention of non-existent entities and the misplacement of temporal facts; concurrently, how to integrate structured TTP extraction with summarization to cut APS omissions and to retain the full, multi-step procedural logic are also significant. These refinements strengthen the connection between our error analysis and the methodological contributions of future CTI summarization research.

CRedit authorship contribution statement

Wei Peng: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Junmei Ding:** Methodology, Data curation. **Wei Wang:** Validation, Data curation. **Lei Cui:** Investigation, Data curation. **Wei Cai:** Validation, Data curation. **Zhiyu Hao:** Validation, Supervision. **Xiaochun Yun:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A.1. Ethical statement

All data in CTISum dataset has been double checked by experts that major in intelligence analysis, and manually checked to ensure the absence of any ethical or political inaccuracies.

A.2. Details of regular expression

During the process of filtering and cleaning the parsed PDF data, in order to reduce the burden of manual checking by annotators, we first sample and observe 50 examples, and then extract some common features to formulate into regular expressions for automatically cleaning the data. Specifically, details of regular expression are in the following:

- Removing consecutive dots (table of contents)
- Deleting redundant newlines/carriage returns
- Removing extra whitespace between characters
- Deleting multiple consecutive lines of IP addresses, hashes, etc.
- Removing website links starting with "http"
- Removing Thai, Korean and other non-English characters

By automating the cleaning of patterns and noise with regex, we aim to improve efficiency and reduce manual effort. The regular expressions are tuned iteratively based on continuously sampling and inspecting. This method allows us to programmatically clean a significant portion of repetitive issues in parsed PDF data before manual review. In the future, we can continue refining the regex and identifying new patterns to clean the data. The goal is to minimize the amount of manual effort while ensuring high-quality cleaned data. Some utilized code of regular expression (Python) can be seen as follows:

```
r'\.{2,}'
r'\{2,}'
r'\s\s+'
r'(\d{1,3}\.){3}\d{1,3}|[a-f0-9]{32,64}'
r'^(\$.+[\n])+|^w+([\^])+\)\s\S{1}\{[\n\r]\s.+'
r'https?:/(?:[-\w.]|(?:%[\da-fA-F]{2}))+'

```

A.3. Hallucination discussion

In this section, we make an analysis to discuss hallucinations between ChatGLM3 and GPT-4o. We categorize hallucinations into three main types: (1) Entity generation or substitution, where the model invents or replaces key entities such as threat actor names, malware families, or CVE identifiers (e.g., misattributing a campaign from APT28 to another group, or replacing a referenced vulnerability with a non-mentioned CVE); (2) Attack-step generation or reordering, where nonexistent intermediate steps are inserted into the attack chain, or correct steps are rearranged in an inconsistent order, which is especially problematic for the APS task; (3) Temporal inconsistencies, where incorrect years or timelines are generated when multiple dates appear in the source document.

Data availability

Data will be made available on request.

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